ASIAN INSTITUTE OF MANAGEMENT

Data Mining and Wrangling

Singular Value Decomposition

Session 19 and 20

BSDSBA 2028

01 April 2026

ASIAN INSTITUTE OF MANAGEMENT

Class Administrative Matters

Course Deliverables

Currently Checking – Midterm Exam

Due this Week – ICA08 (Wed April 1), E6 (Mon April 6), Mini Project 1 (Wed April 1),

MP1 Peer Eval (Mon April 6)

Upcoming – R09 (Wed April 1), ICA09 (Wed April 8), E7 (Fri April 10)

For Checking – ICA04, ICA05, ICA06, E3, R07, E4, E3S2, ICA07, E5, A1, R08

Session 19 and 20 – SVD and LSA

Gameplan

Session 19

Singular Value Decomposition (SVD)

Latent Semantic Analysis (LSA)

Session 20

Applied Example of SVD and LSA

In-Class Activity 9 – SVD and LSA

Singular Value Decomposition (SVD)

Singular Value Decomposition

Prerequisites

Orthonormal Vectors and Orthogonal Matrices

The n vectors $q_1, \dots, q_n$ are ***orthonormal*** if

$$q_i^T q_j = \begin{cases} 0 & \text{when } i \neq j \\ 1 & \text{when } i = j \end{cases}$$

A matrix Q with orthonormal columns has $Q^T Q = I$

Singular Value Decomposition

Prerequisites

Rotation Matrix

An orthonormal matrix Q rotates every vector in a plane by the angle θ:

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Singular Value Decomposition

Example 1: Rotation of a Vector

Multiply the matrix $Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$ with the vector $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Singular Value Decomposition

Prerequisites

Diagonal Matrices

A matrix $\boldsymbol{D}$ is a diagonal matrix if all of it's off-diagonal entries are zero, i.e.,

$$d_{ij} = \begin{cases} 0 & \text{when } i \neq j \\ c_i & \text{when } i = j \end{cases}$$

Singular Value Decomposition

Example 2: Stretching a Vector

Multiply the matrix $\boldsymbol{D} = \begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$ with the vector $\begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$

Singular Value Decomposition

Example 3: Rotating a Vector (again)

Multiply the matrix $\boldsymbol{Q} = \begin{bmatrix} 3/\sqrt{10} & -1/\sqrt{10} \\ 1/\sqrt{10} & 3/\sqrt{10} \end{bmatrix}$ with the vector $\begin{bmatrix} 3\sqrt{10}/2 \\ \sqrt{10}/2 \end{bmatrix}$

Singular Value Decomposition

Example 4: SVD Preview

Singular Value Decomposition

The Core Idea of SVD

The Singular Value Decomposition of any matrix $\boldsymbol{A}$ is

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathbf{T}}$$

$$\boldsymbol{A}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$$

That is, any matrix A can be simplified into a three-step sequence of rotation, stretch, and rotation.

Singular Value Decomposition

The Core Idea of SVD

$$\boldsymbol{AV} = \boldsymbol{U\Sigma}$$

Singular vectors v_i and u_i in $Av_i = \sigma_i u_i$ are orthonormal

$$\boldsymbol{V}^{\boldsymbol{T}}\boldsymbol{V} = \boldsymbol{I}$$

$$\boldsymbol{U}^{\boldsymbol{T}}\boldsymbol{U} = \boldsymbol{I}$$

Singular Value Decomposition

The Core Idea of SVD

$$\boldsymbol{AV} = \boldsymbol{U\Sigma}$$

The diagonal matrix $\boldsymbol{\Sigma}$ contains the singular values σ_i

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$$

Singular Value Decomposition

The Core Idea of SVD

$$\boldsymbol{AV} = \boldsymbol{U\Sigma}$$

The squares σ_i^2 of those singular values are eigenvalues λ_i of $A^T A$ and AA^T.

Singular Value Decomposition

Proof of SVD (Prerequisite)

Recall that for any matrix $\boldsymbol{M}$ with eigenvector matrix $\boldsymbol{W}$ and eigenvalue matrix $\boldsymbol{\Lambda}$

$$\boldsymbol{M}\boldsymbol{W} = \boldsymbol{W}\boldsymbol{\Lambda}$$

$$\boldsymbol{M} = \boldsymbol{W}\boldsymbol{\Lambda}\boldsymbol{W}^{\boldsymbol{T}}$$

Singular Value Decomposition

Proof of SVD

The goal is $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathbf{T}}$

Consider $\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}}$

$$\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{A} =$$

$$\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}} =$$

Singular Value Decomposition

Proof of SVD

The goal is $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathbf{T}}$

Consider $\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}}$

$$\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Sigma}^{\mathbf{T}}\boldsymbol{\Sigma}\mathbf{V}^{\mathbf{T}}$$

$$\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{\mathbf{T}}\boldsymbol{U}$$

Singular Value Decomposition

Proof of SVD

The goal is $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathrm{T}}$

Consider $\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}}$

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$$\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{\mathrm{T}}\boldsymbol{U}$$

$\boldsymbol{V}$ contains orthonormal eigenvectors of

contains orthonormal eigenvectors of $\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}}$

${\sigma_1}^2$ to ${\sigma_r}^2$ are the non-zero eigenvalues of both $\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}}$

Singular Value Decomposition

Proof of SVD (Final Step)

SVD requires $\boldsymbol{AV} = \boldsymbol{U\Sigma}$ or $\boldsymbol{A}v_k = \sigma_k u_k$

Choose v_i such that $\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{A}v_i = {\sigma_i}^2 v_i$ then $u_i = \dfrac{\boldsymbol{A}v_i}{\sigma_i}$

Singular Value Decomposition

Proof of SVD

The goal is $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathrm{T}}$

Consider $\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}}$

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Singular Value Decomposition

The PCA Connection

Consider matrix $\boldsymbol{A}$ such that $\boldsymbol{A}$ is a feature vector matrix $\boldsymbol{D}$ with n rows and d columns. If $\boldsymbol{D}$ is mean-centered such that the covariance matrix is given by $\frac{\boldsymbol{D}^{\boldsymbol{T}}\boldsymbol{D}}{n}$, then the right singular vectors SVD provides the same basis vectors as PCA.

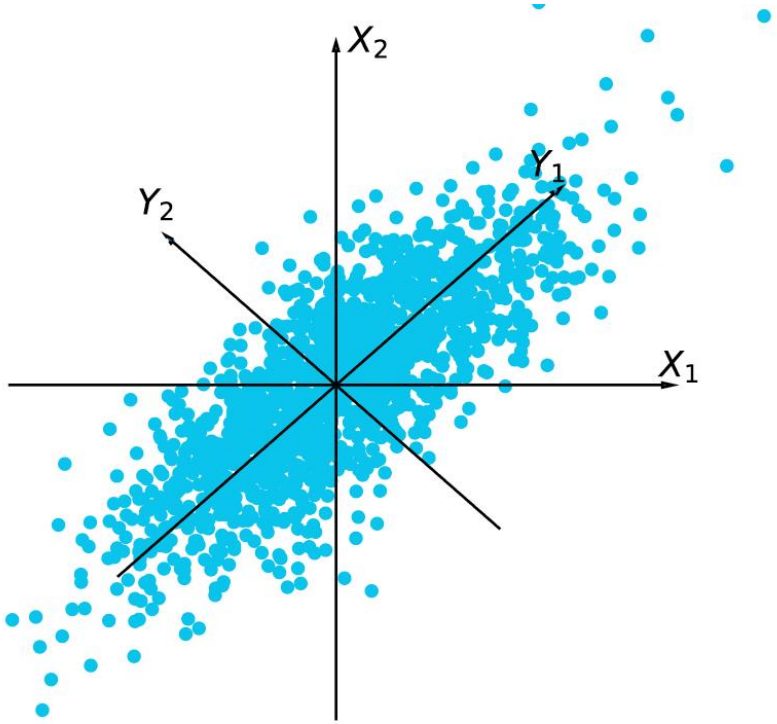
$$\boldsymbol{D}^{\boldsymbol{T}}\boldsymbol{D} = \boldsymbol{V}\boldsymbol{\Sigma}^{\mathbf{T}}\boldsymbol{\Sigma}\mathbf{V}^{\mathbf{T}}$$

and the i-th PCA eigenvalue is given by $\lambda_i = \frac{{\sigma_i}^2}{n}$

Singular Value Decomposition

PCA vs SVD

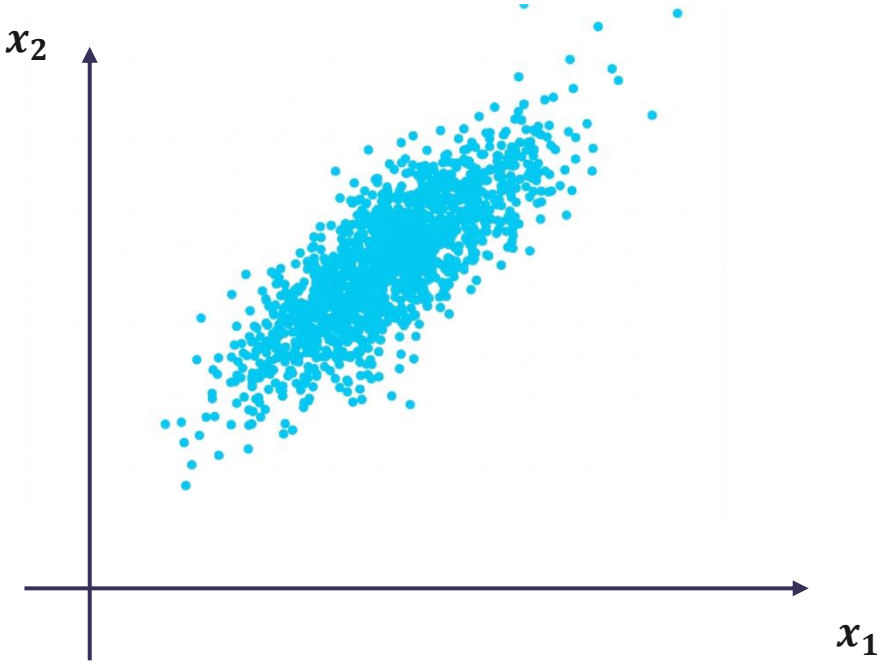
PCA finds a rotated axis-system where the greatest amount of variance is captured in a small number of dimensions with a mean-centered data.



Singular Value Decomposition

PCA vs SVD

SVD finds a rotated axis-system with a maximum aggregated squared Euclidean distance (or *energy*) of the transformed data points about the origin.



Singular Value Decomposition

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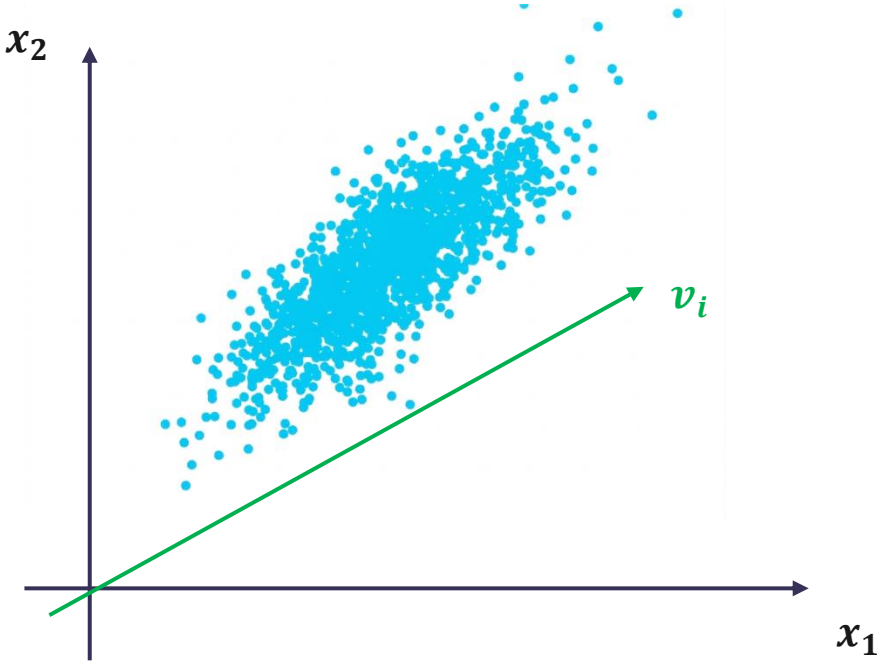

Singular Value Decomposition

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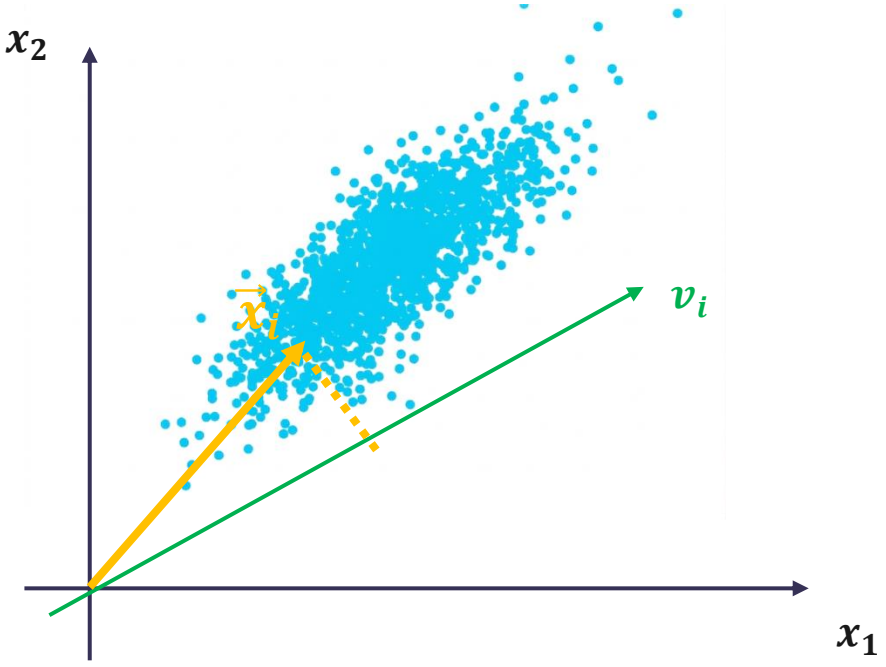

Singular Value Decomposition

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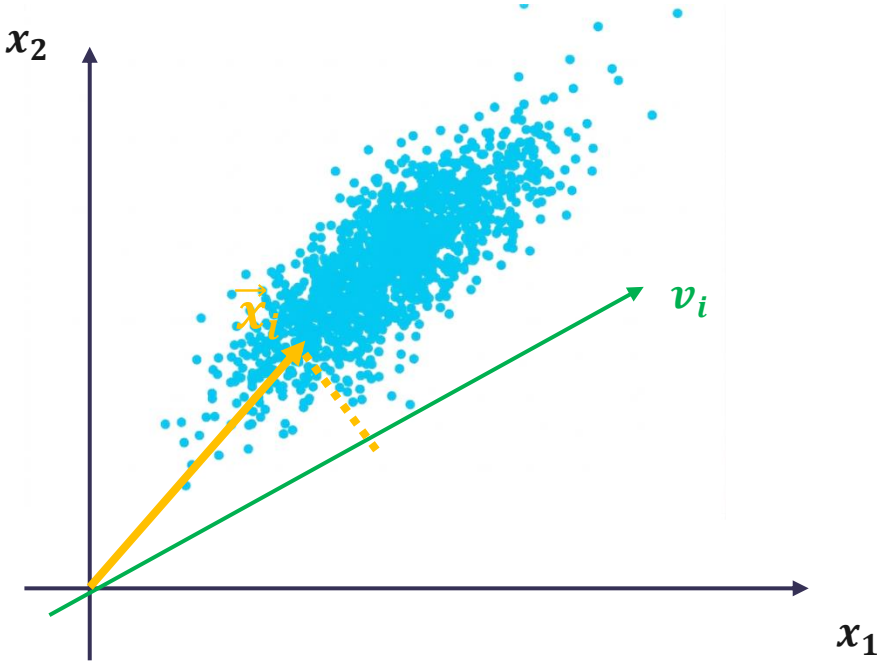
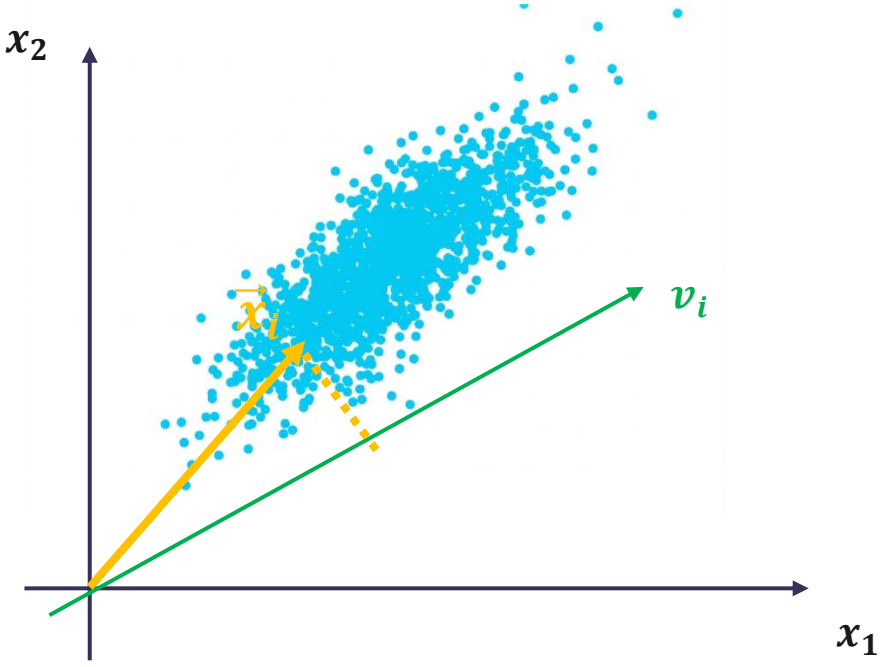


Projected dataset $D\overline{v}$

Singular Value Decomposition

PCA vs SVD

SVD finds a rotated axis-system with a maximum aggregated squared Euclidean distance (or *energy*) of the transformed data points about the origin.



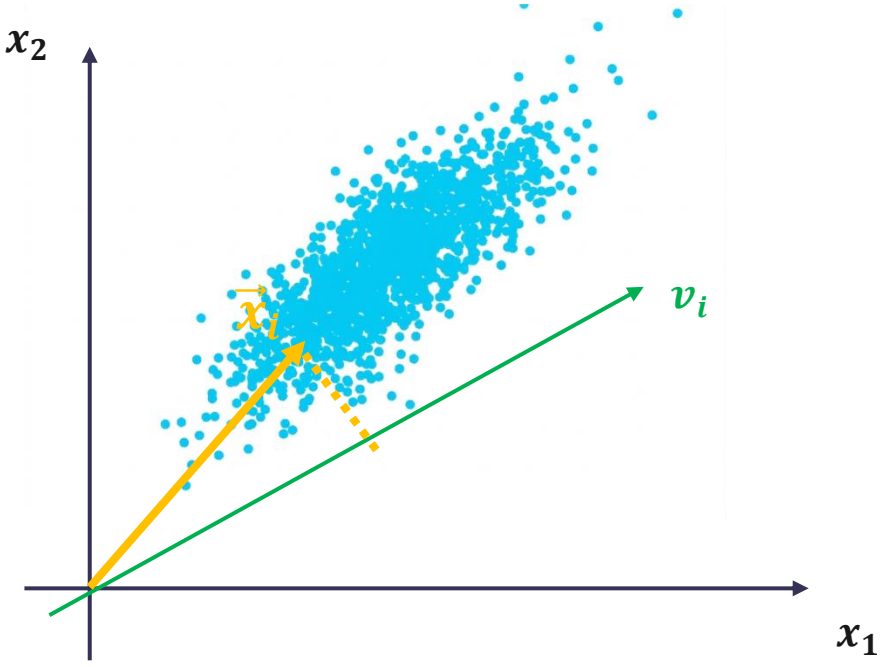
Projected dataset $D\overline{v}$

Maximize $(D\overline{v})^T(D\overline{v})$

Singular Value Decomposition

PCA vs SVD

SVD finds a rotated axis-system with a maximum aggregated squared Euclidean distance (or *energy*) of the transformed data points about the origin.



Projected dataset $D\overline{v}$

Maximize $(D\overline{v})^T(D\overline{v})$

$$\overline{v}^T D^T D\overline{v} - \lambda(\|v\| - 1)^2$$

Singular Value Decomposition

Truncated SVD

When eigenvectors (right singular vectors) with the k largest eigenvalues are selected, the data matrix reduces to:

$$\boldsymbol{D}_{\boldsymbol{k}}' = \boldsymbol{D}\boldsymbol{V}_{\boldsymbol{k}} = \boldsymbol{U}_{\boldsymbol{k}}\boldsymbol{\Sigma}_{\boldsymbol{k}}$$

The total preserved energy in the data projection along the top-k singular vectors is equal to $\sum_{i=1}^{k} {\sigma_i}^2$

Singular Value Decomposition

Interpreting the Singular Vectors

Coordinates – Draw arrows that show the original feature vector coordinates direction with respect to each SVs obtained.

Correlations – Arrows clusters into groups of highly related attributes.

Clusters – Describe the clusters of point observed with respect to the SVs

Funnels – Funnels are wide at one end and narrow at the other. They happen when one SV affects the variance of another SV.

Voids – Areas which are unpopulated.

Singular Value Decomposition

Limitations of SVD

- **SVD** cannot capture nonlinear relationships.
- A core assumption of **SVD** is that high variance means important information.
- **SVs** can reduce interpretability
- **SVD** is sensitive to scaling
- **SVD** is an unsupervised global method

Latent Semantic Analysis (LSA)

Latent Semantic Analysis

What is Latent Semantic Analysis (LSA)?

Latent Semantic Analysis (LSA) is a **natural language processing (NLP)** technique that uses **singular value decomposition (SVD)** not only to represent meaning of words as vectors, but also the entire meaning of documents.

Latent Semantic Analysis

What is Latent Semantic Analysis (LSA)?

Latent Semantic Analysis (LSA) is a **natural language processing (NLP)** technique that uses **singular value decomposition (SVD)** not only to represent meaning of words as vectors, but also the entire meaning of documents.

You shall know a word by the company it keeps.

J.R. Firth

Latent Semantic Analysis

Illustrative Example

Given the following documents, guess the meaning of the Japanese words:

1 i eat リンゴ every morning

2 she likes to eat バナナ after lunch

3 the 猫 is very cute and fluffy

4 my 犬 loves to play in the park

5 i drink 水 when I am thirsty

6 he drinks コーヒー to stay awake

7 the 猫 sleeps on the sofa all day

8 the犬 runs and plays outside

Latent Semantic Analysis

Illustrative Example

Possible grouping of the words

1 i eat リンゴ every morning

2 she likes to eat バナナ after lunch

3 the 猫 is very cute and fluffy

4 my 犬 loves to play in the park

5 i drink 水 when I am thirsty

6 he drinks コーヒー to stay awake

7 the 猫 sleeps on the sofa all day

8 the犬 runs and plays outside

Latent Semantic Analysis

Illustrative Example

LSA builds something like this under the hood:

	food	drinks	pets
リンゴ	0.4	0.1	0
バナナ	0.4	0.1	0
猫	0	-0.1	0.5
犬	0	-0.1	0.5
水	0.1	0.3	0
コーヒー	0.1	0.3	0

Latent Semantic Analysis

SVD (the math behind LSA)

Given a document-term matrix (e.g. TF-IDF matrix) $\boldsymbol{D}$ with n rows (documents) with d terms (dictionary), the truncated matrix $\boldsymbol{D}'$ is given by:

$$\boldsymbol{D}'_{(n\times k)} = \boldsymbol{D}_{(n\times d)}\boldsymbol{V}_{(d\times k)} = \boldsymbol{U}_{(n\times k)}\boldsymbol{\Sigma}_{(k\times k)}$$

Latent Semantic Analysis

SVD (the math behind LSA)

Right Singular Vectors $\boldsymbol{V}_{(\boldsymbol{d}\times\boldsymbol{k})}$

The V matrix contains the term-topic matrix that tells you about “the company the word keeps”. It contains all topic vectors for each word in your corpus as columns.

Latent Semantic Analysis

SVD (the math behind LSA)

Singular Values $\Sigma_{(k \times k)}$

The singular values tell you how much information is captured by each dimension in the semantic (topic) vector space.

Latent Semantic Analysis

SVD (the math behind LSA)

Left Singular Vectors $U_{(n \times k)}$

The matrix U contains the shared meaning between documents, since it measures how often documents use the same topics in the semantic model of the documents.

Latent Semantic Analysis

The Typical LSA Pipeline

1. **Preprocess the text data** – tokenize, case-fold, stemming/lemmatization, stopwords removal, word vectorization (via BoW or TF-IDF).
2. **Perform SVD** – find the left and right singular vectors and singular values.
3. **Decide how many singular vectors to retain** – choose based on the variance explained or total energy preserved, or reconstruction error.
4. **Transform the data using the singular vector matrix** – transform the document-term matrix data to the reduced semantic vector space representation.
5. **Visualize and interpret the singular vectors** – understand what the singular vectors mean and what topics they represent.
6. **Perform downstream analytics tasks** – use the semantic vectors to perform downstream task such as IR or clustering.

AIM